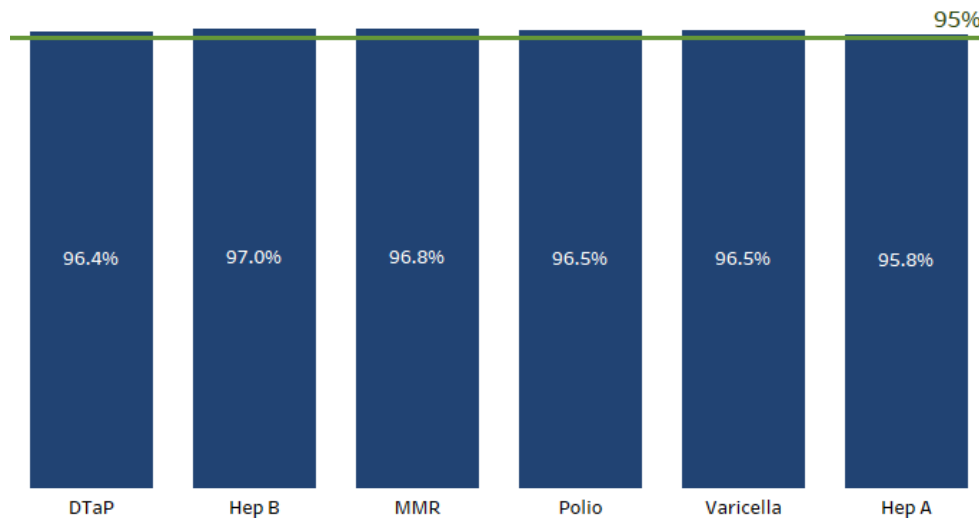


If a student had a valid temporary certificate, a medical exemption, or an incomplete immunization record, schools were asked to report which individual vaccine series the student had completed. Figure 1 shows the percentage of public and private school students who were complete for each vaccine series. Note: The percentages in Figure 1 reflect the sum total of public and private school kindergarten students who were complete for an individual vaccine series, regardless of whether or not they had received all of the vaccines required for school enrollment.

Figure 1. Percentage of public and private school kindergarten students who were complete for each vaccine series, compared to Healthy People 2020 goal of 95% complete



Public Schools Statewide Summary

All 1,016 public schools that were asked to complete the 2019-2020 Kindergarten Assessment submitted results. Statewide, 95% of public school students were fully immunized at the time of the survey (72,737 of 76,588 students). Of all public school students, 67.3% (n=51,562) had their immunization record completed in, and validated by, TennIIS. Students who were not fully immunized fell into the following categories, in decreasing order of prevalence:

- Temporary Certificate: 2.0% (n=1,515)
- Religious Exemption: 1.8% (n=1,399)
- Incomplete Record: 0.7% (n=557)
- Missing Record: 0.3% (n=217)
- Medical Exemption: 0.1% (n=93)
- Transfer <30 days: 0.1% (n=70)

Private Schools Statewide Summary

A total of 182 private schools (out of 255 schools; 71.4%) with kindergarten enrollment completed the 2019-2020 Kindergarten Assessment. This completion rate is higher than in the previous three school years (range: 42.3% to 56.3%). Statewide, 93.8% of private school kindergarten students were fully immunized at the time of the survey (3,757 of 4,007 students). The 2019-2020 Kindergarten Assessment did not ascertain how many private school students had immunization certificates that were validated by TennIIS. Students who were not fully immunized fell into the following categories, in decreasing order of prevalence:

- Religious Exemption: 2.7% (n=110)
- Incomplete Record: 2.1% (n=84)
- Missing Record: 0.8% (n=34)

- Temporary Certificate: 0.3% (n=14)
- Medical Exemption: 0.2% (n=8)
- Transfer <30 days: N/A for private schools based on data collection methods

Trends over Time

Over the past six years, the average annual kindergarten enrollment in TN was 78,500 students (range: 76,181 to 80,595). Annually, approximately 96% of all kindergarten students are enrolled in public schools, and approximately 4% are enrolled in private schools. The percentage of public and private school students who received all vaccines per Tennessee's state requirements for kindergarten entry (i.e., were "fully immunized") has consistently been approximately 95% over the past six years (range: 94.8% to 95.6%). For a community to be at low-risk for a vaccine-preventable disease outbreak, this rate should remain above 95%.

Figure 2 shows the percentage of public school students fully immunized compared to the percentage of students documented as fully immunized in TennIIS (i.e., the child had all his/her required immunizations for kindergarten entry recorded in TennIIS). The percentage of public school students documented in TennIIS rose from 43.4% in the 2015-2016 school year to 67.3% in the 2019-2020 school year; a 24% increase in just five years. This significant increase is attributed to the concerted efforts of the VPDIP program to increase TennIIS participation among immunizing providers and school users statewide.

Figure 2. Percentage of public school students fully immunized, and percentage of public school students whose immunization records were complete in TennIIS, over time

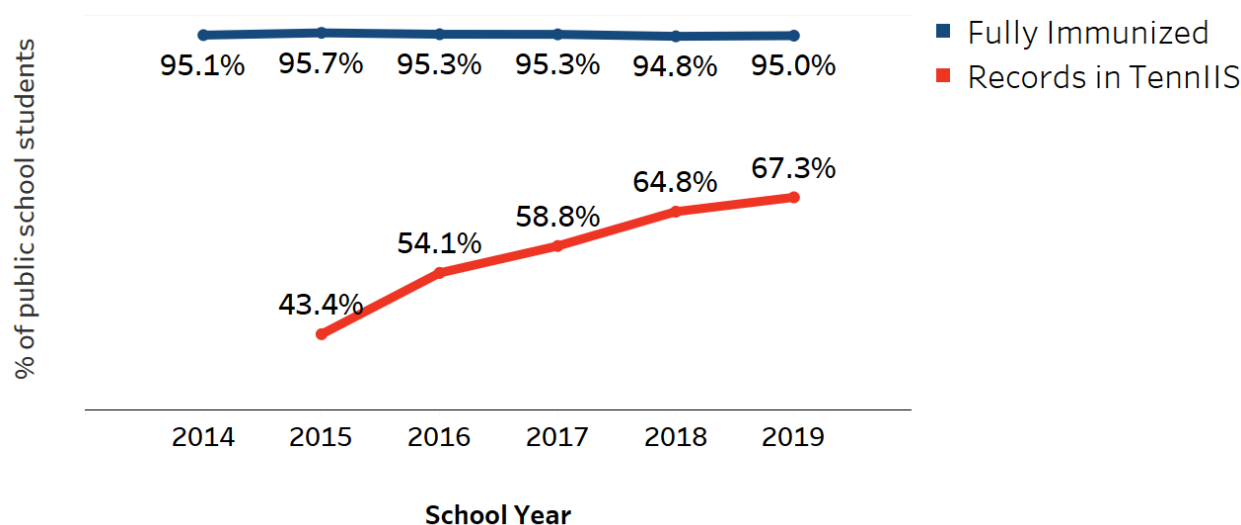
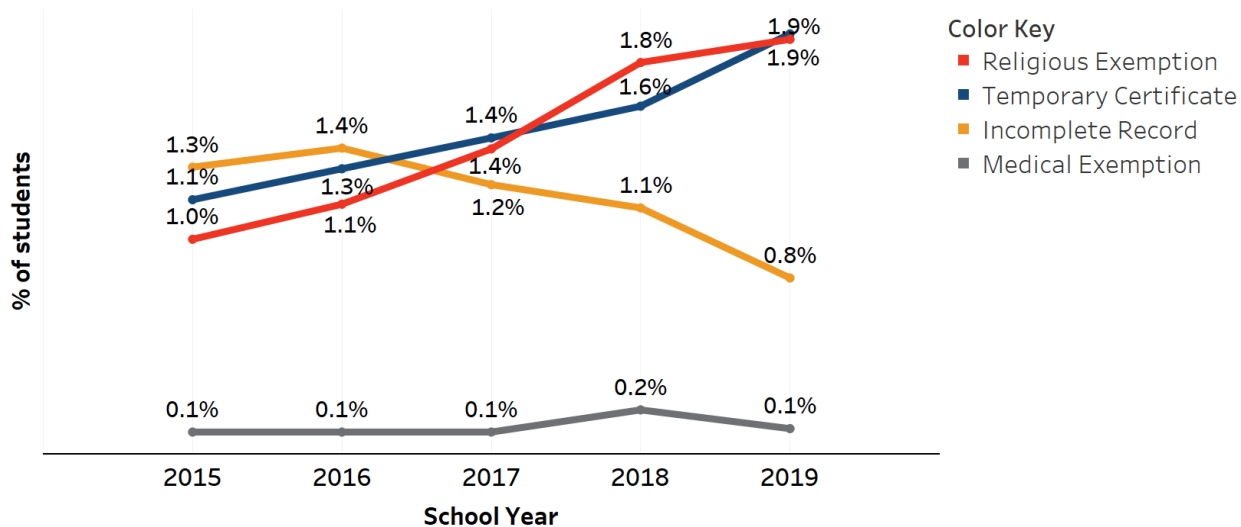


Figure 3 shows the percentage of all TN kindergarten students who were not fully immunized, by category, over time. Over the past five years, the percentage of students with an incomplete immunization record has decreased over time (see Figure 3, gold line). However, the percentage of students claiming a religious exemption and the percentage of students with a temporary immunization certificate have each increased over the past five years by nearly 1% (see Figure 3, blue and red lines). The percentage of students claiming a religious exemption rose from 1.0% to 1.9% between the 2015-2016 and the 2019-2020 school years, and the percentage of students with a temporary immunization certificate rose from 1.1% to 1.9% in the same time frame. It should be noted that none of the major Christian denominations, including Roman Catholic, Baptist, Methodist, Episcopalian, Presbyterian, Lutheran, Pentecostal, Church of Jesus Christ of Latter-Day Saints, Eastern Orthodox, Jehovah's Witness, Mennonite, Quaker, and Unitarian-Universalist object to immunizations. Similarly, Judaism, Islam, Hinduism, Jainism, and Scientology also have no objection to immunizations. The rising

number of religious exemptions in the state of Tennessee is likely due to parents filing a religious exemption in lieu of a personal philosophical exemption, despite the Tennessee Code Annotated stating that doing so is subject to penalty of perjury.⁴ A temporary immunization certificate is issued when a child has not received all required vaccines, but is in the process of completing them. A temporary certificate expires one month after the date that the next required immunization is due, according to the minimum acceptable (“catch-up”) dose interval per the CDC/ACIP schedule.⁵ An expired temporary immunization certificate is not valid proof of immunization; students with an expired temporary certificate are classified as having an incomplete vaccine record in the Kindergarten Assessment.

The percentage of kindergarten students claiming medical exemptions from a vaccine remained stable at approximately 0.1% each year (see Figure 3, grey line). The percentage of students with missing vaccine records over the same time period has also remained stable at approximately 0.5% (not shown in figure).

Figure 3. Percentage of all TN kindergarten students not fully immunized, by category, over time



We hope you find the new format of the 2019-2020 Kindergarten Assessment report useful in understanding the statewide trends in compliance with kindergarten immunization requirements.

Questions and comments regarding this report or the Kindergarten Assessment may be directed to School.Imm@tn.gov.

Sincerely,

Michelle Fiscus, MD FAAP
 Medical Director
 Tennessee Vaccine-Preventable Diseases and Immunization Program
 Tennessee Department of Health

⁴ <https://www.vumc.org/health-wellness/news-resource-articles/immunizations-and-religion>

⁵ <https://www.cdc.gov/vaccines/schedules/hcp/imz/catchup.html>

High immunization coverage rates are critical to avoiding Vaccine-Preventable Disease (VPD) outbreaks (e.g., measles). A VPD outbreak may occur when community members are inadequately immunized against a disease. To remain at low risk for a VPD outbreak the immunization rate should remain at or above 95%. In the maps above and tables below, an immunization rate of at least 95% meets this immunization goal, an immunization rate of 90% to 94.9% needs improvement, and an immunization rate below 90% makes the community members vulnerable to preventable outbreaks.

For detailed Kindergarten Assessment results at the county, school districts, and school levels, please see the *Public School Districts (SD) by County Summary* and the *Private Schools by County Summary* sections below.

Table 1. Percentage of **public school** students fully immunized, total # students, and the number of schools with students fully immunized within each color category, by county, in descending order by “% students fully immunized” (Color Key: <90%; 90-94%; 95-100%)

County	% Students Fully Immunized in County	Total # Students in County	# Schools with <90% students fully immunized	# Schools with 90-94% students fully immunized	# Schools with 95-100% students fully immunized
CLAY	100.0%	85	0	0	2
CHESTER	99.6%	233	0	0	3
OVERTON	98.7%	227	0	1	4
GRAINGER	98.7%	224	0	0	4
STEWART	98.6%	142	0	0	2
HANCOCK	98.5%	68	0	0	1
TIPTON	98.4%	770	0	0	7
CARROLL	98.4%	309	0	0	5
MEIGS	98.3%	118	0	0	2
BLED SOE	98.2%	109	0	0	3
CHEATHAM	97.8%	458	0	0	6
DECATUR	97.8%	135	0	0	2
MARION	97.8%	314	1	0	4
DEKALB	97.8%	223	0	0	2
CROCKETT	97.7%	221	0	1	4
LAUDERDALE	97.7%	259	0	0	2
SCOTT	97.7%	344	1	0	5
CAMPBELL	97.7%	384	0	1	7
DYER	97.6%	463	0	0	6
PERRY	97.6%	83	0	0	2
UNICOI	97.5%	159	0	1	3
HENDERSON	97.4%	387	0	1	7
RHEA	97.4%	384	0	0	5
OBION	97.3%	330	0	1	5
BENTON	97.1%	140	1	0	2
TROUSDALE	97.1%	103	0	0	1
CARTER	97.0%	504	0	3	9
WEAKLEY	96.9%	325	0	0	5

County	% Students Fully Immunized in County	Total # Students in County	# Schools with <90% students fully immunized	# Schools with 90-94% students fully immunized	# Schools with 95-100% students fully immunized
ANDERSON	96.8%	853	0	3	13
HOUSTON	96.8%	94	0	0	2
FRANKLIN	96.8%	372	2	0	6
CUMBERLAND	96.7%	520	0	2	7
SEQUATCHIE	96.6%	148	0	0	1
COCKE	96.6%	408	0	2	8
WARREN	96.6%	464	0	3	5
MACON	96.5%	369	0	0	3
WASHINGTON	96.5%	1,271	1	2	16
LEWIS	96.5%	141	0	0	1
HUMPHREYS	96.4%	224	0	1	2
MARSHALL	96.4%	387	0	1	2
SMITH	96.4%	276	0	2	3
LINCOLN	96.4%	357	0	1	5
GIBSON	96.3%	668	1	2	7
MORGAN	96.3%	187	1	1	3
VAN BUREN	96.2%	53	0	0	1
MCMINN	96.1%	593	0	4	6
HENRY	96.0%	329	0	2	2
COFFEE	96.0%	705	0	4	8
HAMBLEN	96.0%	753	1	2	8
MOORE	96.0%	75	0	0	1
HARDEMAN	96.0%	274	1	2	3
WILSON	96.0%	1,742	0	4	13
DICKSON	96.0%	597	0	4	6
LAKE	95.9%	49	0	0	1
MCNAIRY	95.8%	310	0	0	5
MONROE	95.8%	500	0	1	5
HAWKINS	95.8%	471	2	5	5
SULLIVAN	95.7%	1,475	1	7	16
SUMNER	95.7%	2,267	2	7	18
LOUDON	95.6%	568	0	2	5
GILES	95.6%	295	1	0	3
HARDIN	95.6%	271	0	2	3
FAYETTE	95.6%	293	0	1	3
JOHNSON	95.4%	152	1	0	4
GREENE	95.4%	690	1	4	8
RUTHERFORD	95.2%	4,356	1	12	25
ROANE	95.2%	457	0	3	3

County	% Students Fully Immunized in County	Total # Students in County	# Schools with <90% students fully immunized	# Schools with 90-94% students fully immunized	# Schools with 95-100% students fully immunized
CLAIBORNE	95.2%	310	1	3	3
BEDFORD	95.1%	635	1	1	6
SHELBY	95.0%	12,451	19	35	100
LAWRENCE	95.0%	541	0	3	5
JEFFERSON	95.0%	516	1	3	4
MADISON	94.8%	1,001	1	5	7
WILLIAMSON	94.7%	3,192	5	9	19
MONTGOMERY	94.5%	3,100	1	12	10
HICKMAN	94.5%	293	0	1	1
PUTNAM	94.5%	892	1	5	4
KNOX	94.5%	4,186	6	18	26
BLOUNT	94.0%	1,412	2	11	6
ROBERTSON	94.0%	846	1	4	5
WAYNE	93.9%	164	0	2	1
POLK	93.9%	164	1	1	2
MAURY	93.7%	1,016	0	11	3
WHITE	93.5%	310	2	1	3
SEVIER	93.5%	1,119	1	8	4
GRUNDY	93.3%	135	2	2	2
BRADLEY	93.3%	1,162	4	4	9
DAVIDSON	93.0%	7,048	21	29	35
CANNON	92.9%	155	2	1	3
HAMILTON	92.0%	3,584	11	14	18
HAYWOOD	90.3%	217	0	1	0
PICKETT	89.8%	49	1	0	0
JACKSON	88.0%	100	2	0	0
UNION	85.0%	273	3	3	0
FENTRESS	84.7%	202	2	2	0

Table 2. Percentage of **private school** students fully immunized, total # students, and the number of schools with students fully immunized within each color category, by county, in descending order by “% students fully immunized” (Color Key: <90%; 90-94%; 95-100%)

County	% Students Fully Immunized in County	Total # Students in County	# Schools with <90% students fully immunized	# Schools with 90-94% students fully immunized	# Schools with 95-100% students fully immunized
BEDFORD	100.0%	17	0	0	1
DYER	100.0%	13	0	0	1
LINCOLN	100.0%	13	0	0	1
MACON	100.0%	8	0	0	1
TIPTON	100.0%	26	0	0	1
WARREN	100.0%	12	0	0	1
SHELBY	97.6%	889	2	4	31
SUMNER	97.6%	41	0	1	3
WILSON	97.2%	108	1	0	5
MAURY	97.1%	69	0	2	1
HARDIN	94.7%	19	1	0	1
DAVIDSON	93.9%	969	8	5	20
PUTNAM	93.8%	16	0	1	0
MONTGOMERY	93.1%	87	2	0	3
HAMILTON	93.1%	390	2	3	9
HAMBLEN	92.7%	55	0	1	1
SULLIVAN	92.3%	13	0	1	0
WILLIAMSON	91.8%	171	4	3	3
FAYETTE	91.7%	36	1	0	1
ROBERTSON	91.6%	83	2	2	1
KNOX	90.6%	521	8	2	16
RUTHERFORD	90.2%	193	3	1	4
MADISON	90.0%	50	1	0	1
WASHINGTON	89.6%	77	2	1	2
BRADLEY	88.9%	18	2	0	1
CHEATHAM	88.2%	17	1	0	0
BLOUNT	87.9%	33	1	1	0
ANDERSON	85.7%	21	1	0	0
SEVIER	85.7%	35	2	0	0
GREENE	57.1%	7	1	0	0

Public School Districts (SD) by County Summary

The following series of tables and graphs capture information at two different levels: 1) at the county-level, featuring information for all school districts within the county; and 2) at the school district level, featuring information for all schools within the school district. Data are presented by county in alphabetical order, with school districts nested under each respective county in alphabetical order, and schools nested under each respective school district in alphabetical order.

Below are descriptions of each category shown in the county and school district tables and graphs. All table percentages on a given row are the percentage of the “Total Students” number on that row.

County	Tennessee county name. School districts are grouped into their corresponding county.
School District	Department of Education school district name. Schools are grouped into their corresponding school district.
School Name	School name, listed alphabetically within each school district.
Total Students	Total number of enrolled kindergarten students at the time of immunization record assessment.
Fully Immunized	Number and percentage who meet state school entry requirements by being fully immunized.
Religious Exemption	Number and percentage with a valid religious exemption.
Medical Exemption	Number and percentage with a valid medical exemption to one or more vaccines.
Temporary Certificate	Number and percentage with a valid temporary certificate.
Excluded Transfer	Students exempt from assessment because records are unavailable due to a recent transfer from another TN school (within 30 days of the assessment).
Incomplete Record	Number and percentage who do not meet requirements because they have an incomplete vaccination record.
Missing Record	Number and percentage who do not meet requirements because the school has no official immunization certificate on record.